

5 Dynamic Tracing with DTrace



dtrace command in a script. The -s option means compile the D program source file.

```
#!/usr/sbin/dtrace -s
/* The above line means that the script that follows
   needs to be interpreted using dtrace.
   D uses C-style comments.
*/
pid1234:::entry
{}
```

Replace the 1234 in the above command or script with the id of a process that you own that you want to examine. Remember to make your script file executable. Use Ctrl-C to return to the shell prompt.

You should see output similar to the following:

```
# dtrace -n pid23:::entry
dtrace: description 'pid23:::entry'
matched 7954 probes
CPU      ID
  1 105673
  1 105912
^C
#
```

Quick Tip
You can find the pid of a program by running pgrep command. So if you want to know the pid of firefox, then open a terminal and run \$ pgrep firefox and you will get the pid

On systems other than Solaris systems, you must be the super user on the system to execute the dtrace command or D scripts. On Solaris systems, you must be the super user on the system to use DTrace, or you must have appropriate DTrace privileges. You must be the root user to profile or trace processes owned by root. To grant DTrace privileges, become the super user and use the usermod command for the specified non-root user or edit the /etc/user_attr file directly. You should only need dtrace_user and dtrace_proc privileges.

To make the above script easier to reuse, modify the script to take the process id as a parameter, as shown below.

```
#!/usr/sbin/dtrace -s
pid$1:::entry
{}
```